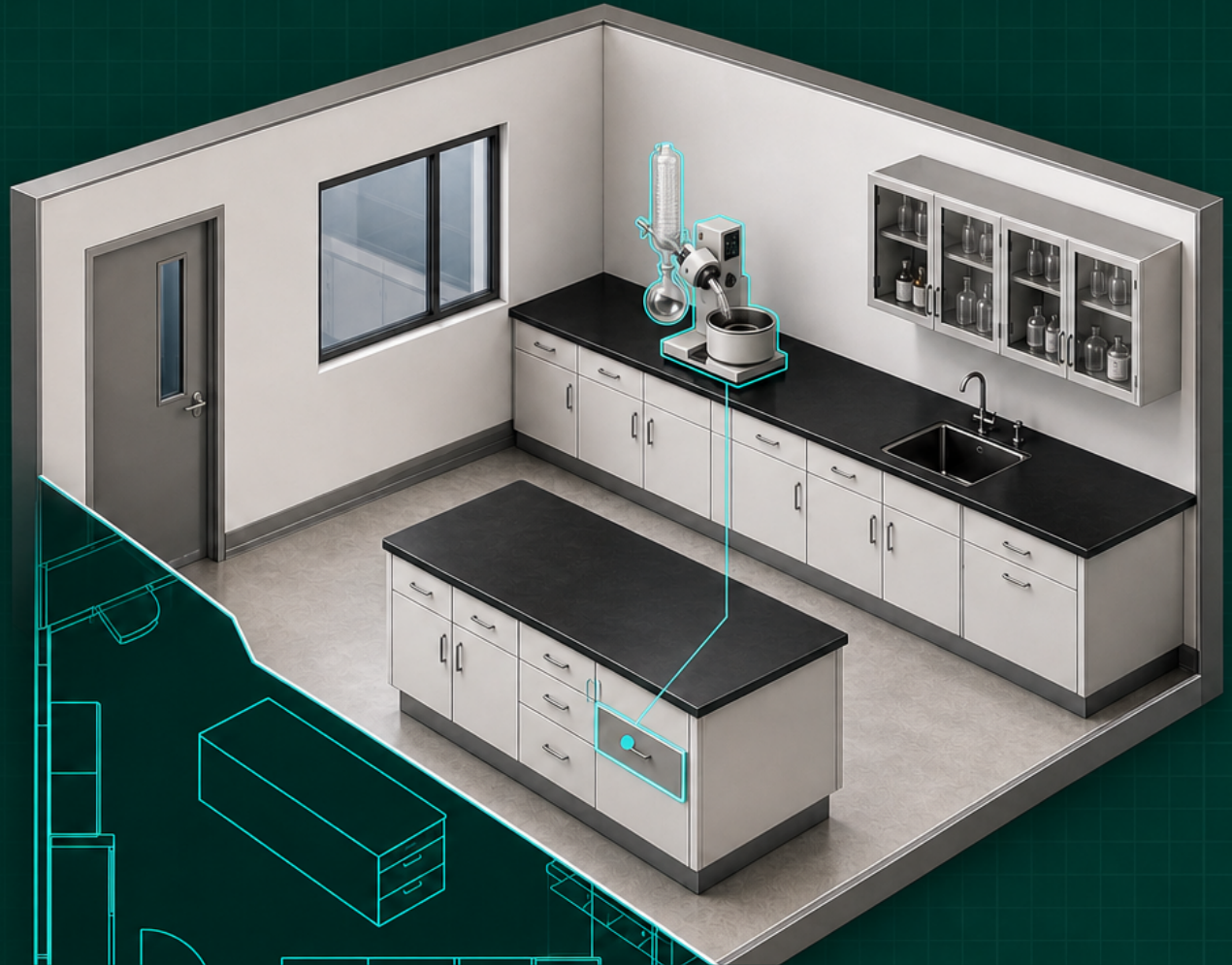


LabSpace AI

DESIGN • INDEX • FIND



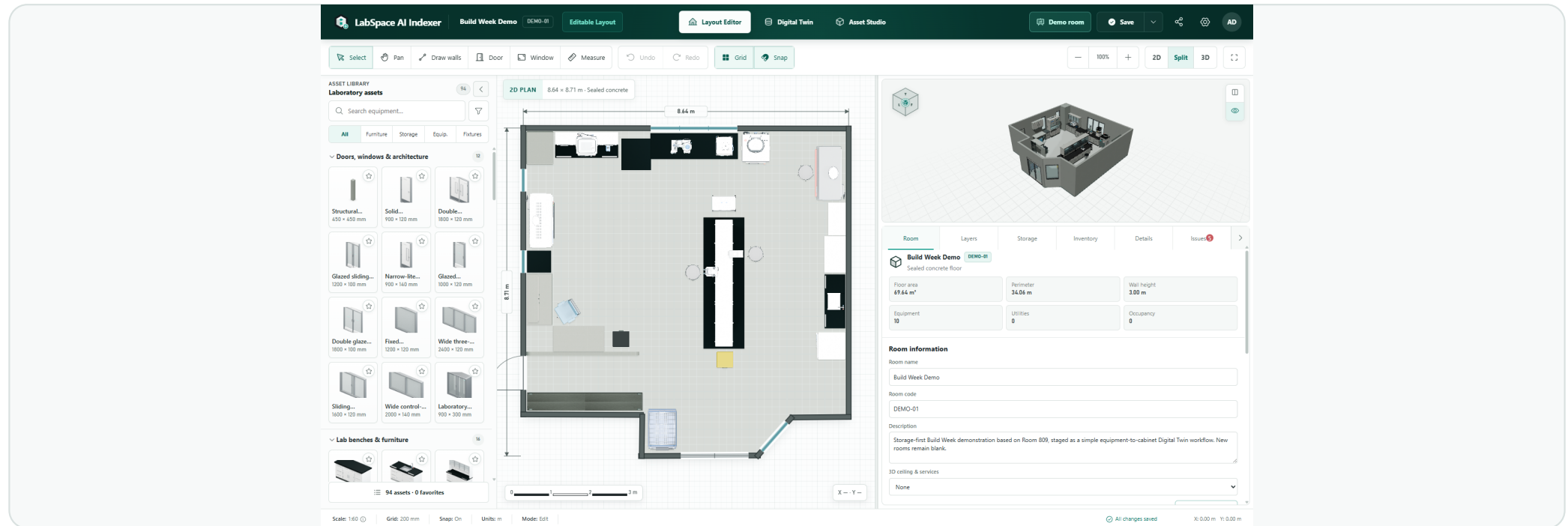
WORK & PRODUCTIVITY

LabSpace AI: From Layout Builder to Intelligent Spatial Digital Twin

Design the lab. Index every asset. Find anything instantly.

A professional laboratory Layout Builder

The starting point: one spatial model for designing rooms before connecting operational evidence.



THE PROBLEM

Laboratory floor plans, inventory lists, cabinet records, photographs, spreadsheets, and staff knowledge often live in separate systems. The same equipment can be visible in a room yet hard to find operationally.

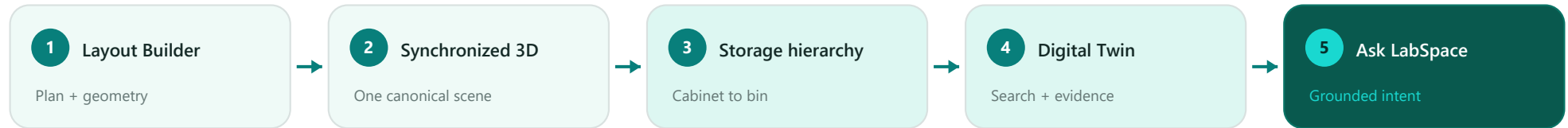
LabSpace began by making the room itself editable, measurable, and trustworthy.

LAYOUT FOUNDATION

- Synchronized material-aware 2D plan and orbitable 3D room.
- Walls, hosted openings, room surfaces, equipment placement, and direct manipulation.
- One millimetre-based scene drives selection, persistence, and deterministic spatial validation.
- Realistic authored assets use the same geometry in the room, plan, and catalog.

From room design to grounded spatial operations

Competition work connected layout geometry, nested storage, evidence, and natural-language navigation.



BEFORE BUILD WEEK

Existing prototype foundation

- Local laboratory editor prototype.
- Versioned canonical scene schema.
- Early procedural planning-asset catalog.

Scope boundary

The original work established the editor and data foundation. The competition contribution focused on making the complete spatial-digital-twin loop coherent, testable, and judgeable.

ADDED OR SUBSTANTIALLY REBUILT

Competition implementation

- Blank-start multi-laboratory rooms and user-owned demo persistence.
- Camera-preserving synchronized 2D/3D editing, split resizing, continuous walls, hosted openings, elevation, flips, and direct wall editing.
- Exact cabinet, shelf, drawer, compartment, and bin hierarchies with stable codes.
- Project-wide Digital Twin search, evidence photographs, exact-location camera focus, and opt-in access previews.
- Ask LabSpace grounded local orchestration over canonical records and deterministic validator results.
- Authored GLBs, offline decoder delivery, regression coverage, release checks, screenshots, and judge documentation.

PUBLIC REPOSITORY EVIDENCE

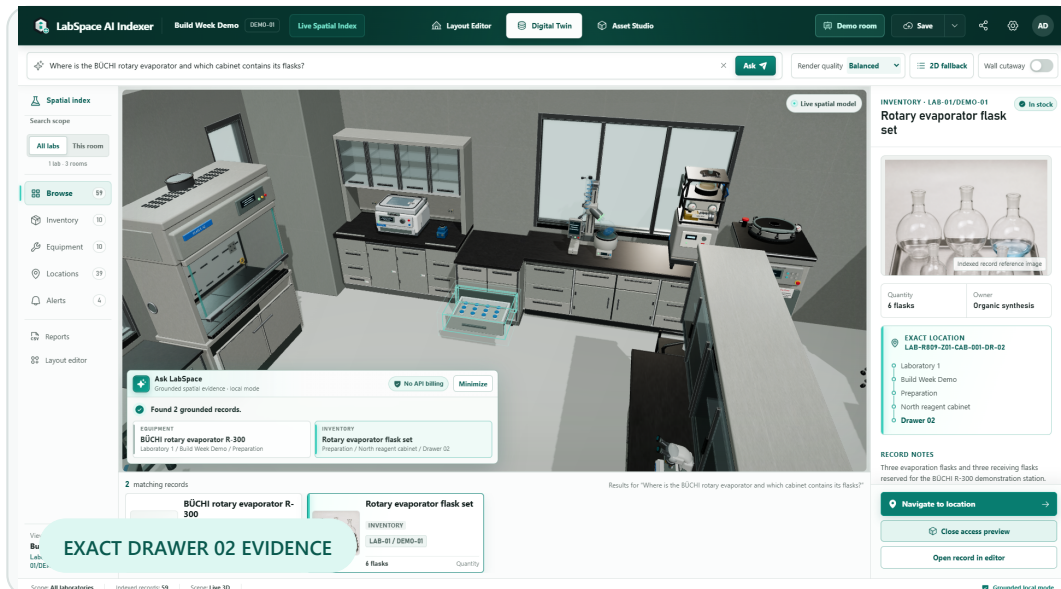
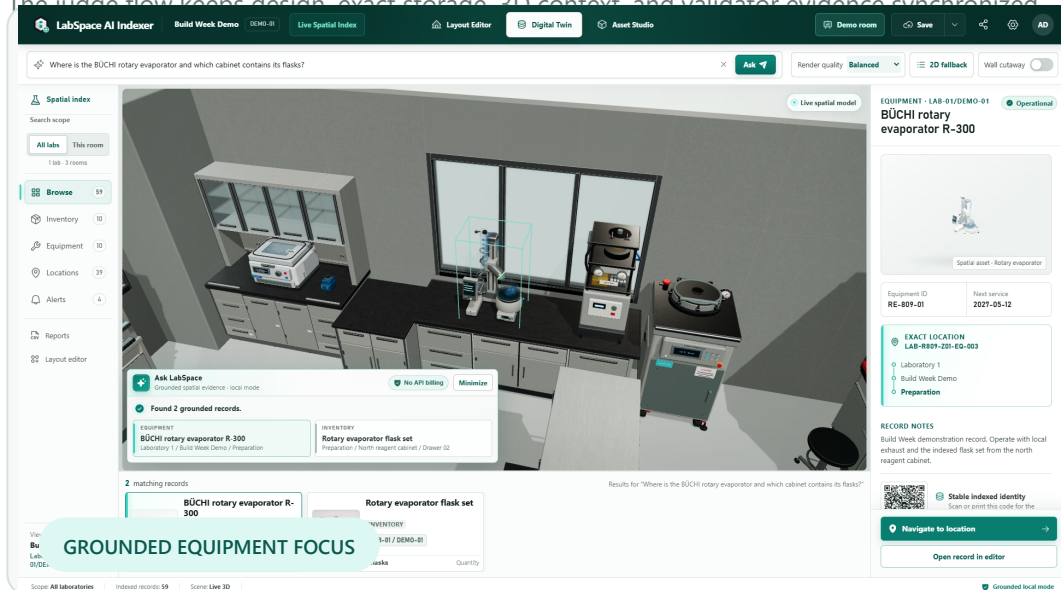
98fbca1 complete competition build

639a76d finalized product screenshots

Current release gate: PASS

Ask, focus, move, validate

The judge flow keeps design, exact storage, 3D context, and validator evidence synchronized.



Complete judge flow

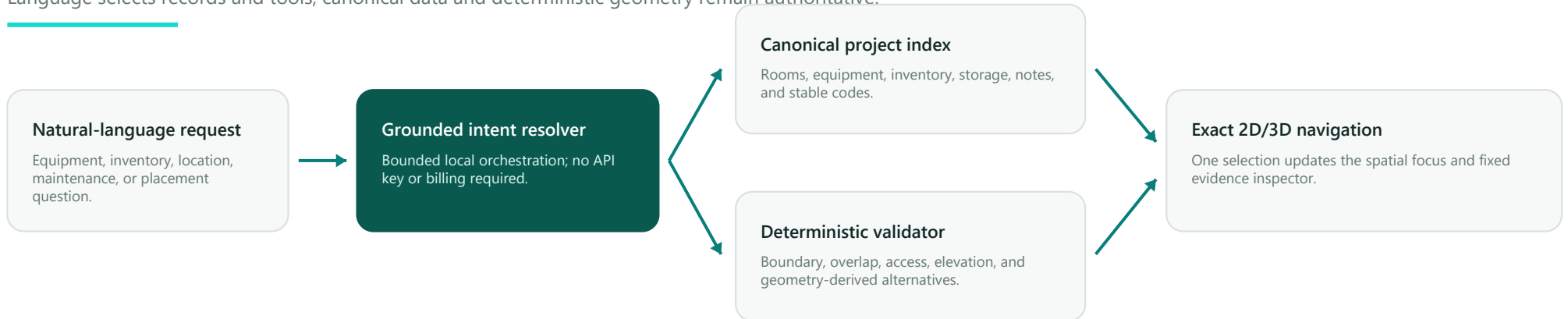
- 1 Design the room**
Draw walls, set materials, and inspect the synchronized plan and 3D room.
- 2 Place equipment**
Use authored planning assets and deterministic placement feedback.
- 3 Index physical storage**
Create cabinet, shelf, drawer, compartment, and bin records with stable codes.
- 4 Ask the location question**
Request the BÜCHI rotary evaporator and the cabinet containing its flasks.
- 5 Focus exact evidence**
Navigate first to the equipment, then North reagent cabinet / Drawer 02.
- 6 Move in 2D**
Update the equipment position without resetting the user's current 3D orbit.
- 7 Synchronize evidence**
The 3D object, indexed record, search result, and location context stay coherent.
- 8 Explain placement**
Report the actual deterministic conflict and offer a geometry-derived valid alternative.

GROUNDED GUARANTEE

Ask LabSpace uses stored records and deterministic validation. Missing evidence remains missing; inventory, ownership, maintenance, utilities, and safety facts are never invented.

A grounded assistant over trusted spatial services

Language selects records and tools; canonical data and deterministic geometry remain authoritative.



96

validated assets

74

authored GLBs

117

passing tests

15

DEMO-01 locations

10

inventory records

10

equipment records

CODEX / GPT-5.6 COLLABORATION

- Architecture and product implementation.
- Spatial-assistant and deterministic tool design.
- Asset-pipeline scripting and authored-model workflows.
- Debugging, browser verification, regression tests, and release audits.
- Competition UI iteration and submission documentation.

PRIMARY CODEX SESSION

019f6a4d-25a9-7812-804c-88b695589b2a

RUNTIME BOUNDARY

Local deterministic mode

The bundled Ask LabSpace runtime is not a live GPT response. It resolves bounded intents against canonical SQLite-backed project data and delegates placement questions to deterministic validators.

A future hosted GPT-5.6 provider can improve intent selection without replacing trusted spatial records, exact-location evidence, or validation services.

A spatial operating system for laboratory work

Measured facts stay separate from potential impact and future integrations.

INTENDED USERS

Laboratory teams

Researchers

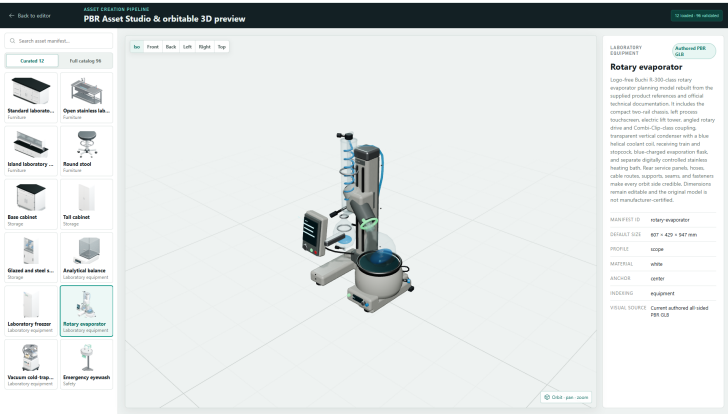
Facility managers

Inventory administrators

Safety and planning staff

POTENTIAL IMPACT - NOT YET MEASURED

Potential benefits include faster search-to-location, clearer handoffs, and fewer disconnected records. No productivity or safety outcome is claimed without a measured study.



RESOURCE-CONSCIOUS COMPETITION PATH

I did not receive the promotional Build Week Codex credits. Development therefore relied on my existing ChatGPT subscription and a resource-conscious workflow. This influenced the decision to make the competition demo fully testable without API billing through a grounded local orchestration layer. A hosted GPT-5.6 provider can later be added without replacing the trusted spatial-data and validation services.

FUTURE ROADMAP: OPTIONAL SFFPS INTEGRATION

SFFPS is my separate GPT-5.5/GPT-5.6-assisted laboratory management project. It is not part of this submission and is not required for LabSpace to work. A future integration could connect SFFPS laboratory operations with LabSpace's spatial design, physical indexing, and exact-location digital twin.

Design the lab. Index every asset. Find anything instantly.